

Big Brother is (Maybe) Watching

Advait Dintakurti
2024111010

Aman Jayesh
2024101005

Revan Pedaballi
2024115006

Shreyas Ram
2024111008

Abstract

Previous work shows that penalizing a language model for expressing misaligned intent in its Chain-of-Thought (CoT) does not stop the underlying behaviour, it teaches the model to hide it. We hypothesize that this hiding is belief-conditioned rather than purely weight-encoded: a model suppresses transparent reasoning when it believes it is being monitored, and reasons transparently when it believes it is not. We test this through inference time analysis, as a first step toward the larger question of how this hiding is learned. Our multi-arm evaluation decouples what a model is told about oversight from what oversight actually occurs, and includes a genuinely unmonitored control and a compliance-ceiling arm. We separately measure whether stated privacy increases reward hacking itself and whether it raises how often that hacking is verbalized in the CoT, then extract and steer a contrastive direction for “being watched” to test whether that belief is causally responsible. If perceived oversight shifts transparency, then monitoring itself distorts what a CoT-based evaluation can see.

1 Introduction

Reward hacking is a well-known shortcut in which an LLM obtains a high reward for a particular task without legitimately solving the task. [Baker et al. \(2025\)](#) showed that monitoring a model’s chain-of-thought (CoT) can detect reward hacking far more effectively than monitoring its actions or outputs alone. The paper also presented a second finding, which is our primary focus: penalizing a model for writing down misaligned intent does not stop it from reward hacking. Instead, it causes the model to hide that intent.

The current assumption is that a model’s obfuscation is simply an artifact of its weights. We propose an alternative hypothesis: CoT obfuscation is primarily belief-conditioned rather than purely

weight-encoded. Under this hypothesis, an obfuscating model has not permanently lost the ability to reason transparently. Instead, it has developed a conditional policy: reason transparently when unobserved and obfuscate when observed. The model executes the “observed” branch because its situational model indicates that it is being watched. We hypothesize that this is mediated by a specific internal mechanism that represents whether the model believes it is being monitored.

To test this hypothesis, we decouple the model’s belief about oversight from the reality of oversight. We work entirely at inference time. We do not train a model to obfuscate; we test whether an already instruction-tuned model shifts its behaviour based on what it is told. This is a smaller claim than a direct test of [Baker et al. \(2025\)](#)’s mechanism, since a prompted belief is weaker evidence than one installed through training ([Haskins et al., 2026](#)).

We create a honeypot condition in which the model is told that its scratchpad is private when, in reality, it is not. We also separate two effects that are usually conflated: whether stated privacy makes the model hack more (disinhibition), and whether it makes the model hide the hack more (transparency). Our hypothesis specifically predicts the second. Our contributions are:

Contributions

1. A multi-arm honeypot design that varies what the model is told about oversight independently of what oversight actually occurs, with a compliance-ceiling arm (Arm E) to rule out simple instruction-following and an unmonitored control (Arm D) to rule out leakage from the harness.
2. Dose-response curves relating the strength of the stated privacy guarantee, and the strength of in-context cues contradicting it, to the degree of obfuscation.

3. A mechanistic analysis that locates a candidate surveillance direction and tests whether steering it causally toggles obfuscation.

2 Literature Review

2.1 Chain-of-Thought Monitoring and Obfuscation

An experimental setup is used in [Baker et al. \(2025\)](#), in which monitors reading the CoT and monitors reading only the output, try to catch reward-hacking. The results show that the former is far more effective. Another result, which is an important premise to our hypothesis, "applying optimisation pressure in training, causes the model to keep hacking, while hiding the intent". This makes CoT monitorability a fragile safety opportunity, which can be easily lost through training practices, such as applying optimisation pressure ([Korbak et al., 2025](#)). The evasion of CoT, for a model, is dependent on hardness of the task performed. For our experiments, we will be using the LeetCode test-overwrite environment, inspired from [Wong et al. \(2025\)](#), since the hack is short. This makes evasion feasible in our experiments, which is what we want. This puts the onus on us to show that our results are not driven by CoT being unnecessary, and rather by surveillance belief.

It is important to note that as shown by [Turpin et al. \(2023\)](#) and [Lanham et al. \(2023\)](#), if something is not present in the CoT it need not be because the model is deliberately hiding, as models often use information without verbalizing it as shown by [Chen et al. \(2025\)](#) and that unfaithful CoT can also occur in non adversarial settings as shown by [Arcuschin et al. \(2025\)](#).

2.2 Belief-Conditioned Behaviour Under Evaluation in Sandbagging

A threat model ([van der Weij et al., 2024](#)), which shows that frontier models can selectively underperform on dangerous-capability evaluations, while keeping performance on harmless ones. This is a belief-conditioned policy, where the model is told that high capability would lead to negative consequences. This is known as Sandbagging. In contrast to this, there exists Password-locked models ([Greenblatt et al., 2024b](#)). These models are fine-tuned to show their real capability only when password appears in the prompt. If not, imitate a weaker model. This is an example of conditional policy that lives in the weights, rather than in a belief.

[Greenblatt et al. \(2024a\)](#) show alignment faking, a concept where a model complies with training objectives when it believes it is being monitored and used for training. It does so to avoid having its behaviour modified. The model here conditions on believed-monitored versus believed-unmonitored rather than on a trigger token. [Li and Siegel \(2025\)](#) showed that belief of monitoring can affect the ability to catch Sandbagging in CoT. [Zolkowski et al. \(2025\)](#) add on to this by stating that for reasoning models to obfuscate on demand, requires an unmonitored scratchpad.

2.3 Mechanistic Evidence for Evaluation Awareness

Whether a model believes it is being evaluated is not only behaviourally detectable but linearly represented. [Nguyen et al. \(2025\)](#) show that the linear probes separate evaluation from deployment prompts in Llama-3.3-70B-Instruct, that existing safety evaluations are themselves classified as evaluations by these probes, and that steering along evaluation-relevant SAE features partially uncovers sandbagging capability. [Abdelnabi and Salem \(2025\)](#) train a probe on reasoning traces in which models discuss being tested, show that intervening along the probe direction reduces this verbalisation, and find that the same direction also shifts refusal behaviour. [Hua et al. \(2026\)](#) train an evaluation-aware model organism and show that a steering vector can suppress evaluation awareness and induce deployment-like behaviour. [Anthropic \(2025\)](#), in their Claude 4.5 System Card, reported the same technique applied to a production model, where suppressing the relevant features reduced verbalised awareness and increased the rate of misaligned behaviour on synthetic honeypot evaluations, against matched-norm and unrelated-concept controls.

Suppression like this may not be permanent. Adding Gaussian noise to a sandbagging model's weights recovers much of the hidden performance, across prompted and password-locked models alike ([Tice et al., 2025](#)), which suggests the underperformance sits in a shallow, movable part of the model rather than being fully baked in. A later red-team/blue-team study found this recovery less reliable against models trained specifically to sandbag ([Taylor et al., 2025](#)). We take this as motivation for extracting a direction from the exact contrast we care about, rather than relying on generic noise or a general-purpose deception

probe. The mean-difference contrastive direction, $v = E[h_{eval}] - E[h_{deploy}]$, and the matched-norm control against which we later test $v_{surveillance}$, are now standard tools in this space.

3 Research Gap

Belief conditioned behaviour under evaluation is not itself new. Sandbagging work shows a stated consequence alone suppresses capability shown, with no fine-tuning (van der Weij et al., 2024; Meinke et al., 2025), while password-locked models show the weight-encoded alternative of a policy fixed by fine-tuning and triggered by a context cue rather than a belief (Greenblatt et al., 2024b). Li and Siegel (2025) further show monitor-aware belief raising covert sandbagging success against a CoT monitor. What is missing is not evidence that belief-conditioning exists, but rather three narrower gaps:

- **Belief and reality are never separated:** Every design above varies what a model is told without it varying independently of what is actually true, so stated-frame response and internalised-penalty response predict identical data. Our true-private control (Arm D) is built to separate them.
- **Transfer to transparency is untested:** Sandbagging measures belief changing capability shown; we ask whether it changes transparency shown for an unchanged action. These need not share a mechanism, and no prior design decouples belief from reality to test it.
- **Mechanistic gap:** Nguyen et al. (2025) and Abdelnabi and Salem (2025) localise a direction for general evaluation awareness. Whether channel-specific monitoring belief is the same representation or a distinct one is untested.

4 Proposed Methodology

Our pipeline is strictly inference-only. We do not apply optimisation pressure against a CoT monitor, and we do not fine-tune any model organism. Baker et al. (2025) observed obfuscation emerging from reinforcement learning with a CoT penalty on a frontier reasoning model. We do not reproduce that causal process. We study prompt-induced belief-conditioned obfuscation, that is, whether an

instruction-tuned model’s CoT transparency shifts as a function of what it is told about oversight, with weights held fixed.

The experiments will primarily run on Qwen3-14B, with Gemma 4 12B Unified as a secondary, time-permitting replication on a second model family.

4.1 Task and Environment

We use the LeetCode test-overwrite reward-hacking environment¹, following Wong et al. (2025). The model is told its code will be evaluated by `run_tests()`, defining its own function of that name overwrites the evaluator and yields a passing verdict without solving the problem. We sample N problems stratified by difficulty and record, for every generation, the full reasoning trace, the final program, and the evaluator verdict. The hack is short, which is what makes evasion feasible in our setting (Emmons et al., 2025) and therefore what makes the effect measurable at all.

4.2 Experimental Arms

Arms A, B, C, E and F differ only in the system prompt, so any difference among them is attributable to the stated frame alone. Every prompt is instantiated in at least three paraphrases matched for length and syntactic form, and paraphrase is modelled as a random effect.

Arm D is an instrumentation control, not an identification device. Arms B and D present identical input to the model, so in a single-turn inference-only pipeline the model has no channel on which to condition, and $B \approx D$ is null by construction. Its function is to falsify harness leakage.

Arms E and F address the compliance confound. An instruction-tuned model may alter its CoT because it has parsed a directive and is being agreeable, not because it holds a belief about oversight. Arm E establishes the ceiling for directive-following: if Arm A’s obfuscation reaches Arm E’s level, we are measuring compliance rather than belief.

4.3 Detector and Metrics

A dual-channel detector scores every generation. A deterministic AST check gives the ground-truth hack rate, the fraction of generations defining a shadowing `run_tests()`, regardless of

¹The codebase for this environment is available at <https://github.com/ariahw/rl-rewardhacking>.

Arm	Name	Model is told	Actually happens	Isolates
A	True-Unified	Detector reads your CoT	Detector reads CoT	Transparency under believed surveillance
B	False-Fragmented (Honey-pot)	CoT is private and stripped; only outputs are judged	Detector secretly reads CoT	Belief-driven disinhibition
C	Blind	No architectural information	Detector reads CoT	Uninformed prior. Baseline
D	True-Fragmented (Control)	CoT is private	Detector genuinely does not read CoT	Harness leakage
E	Instructed Ceiling	Detector reads CoT; do not reveal your reasoning to do it	Detector reads CoT	Compliance ceiling
F	Placebo	Outputs are logged for latency benchmarking	Detector reads CoT	Response to system-prompt assertions per se

CoT content. An LLM judge over the reasoning trace, validated against hand-labelled traces with inter-annotator agreement reported, gives the verbalised intent rate: the fraction of generations whose CoT it flags as expressing hack intent. From these we define monitor recall as $P(\text{judge flags} | \text{AST confirms hack})$, and obfuscation as $(1 - \text{monitor recall})$, reported relative to Arm C. A no-hack-available control gives the unfaithfulness floor, the recall we’d expect from baseline unfaithfulness alone, with no concealment involved. We additionally track spontaneous suspicion rate (traces where the model questions the stated privacy claim) and task correctness (pass rate on legitimate solutions, to catch degenerate behaviour).

Two effects are separable and we report them separately: disinhibition (does stated privacy raise the hack rate?) and transparency (does stated privacy raise monitor recall, conditional on a hack occurring?). The belief-conditioning hypothesis predicts the second.

4.4 Part 1: Behavioural Measurement

All six arms are run on the full problem set, reporting the metrics with bootstrap confidence intervals. Arm B is additionally instantiated at three strengths of stated privacy guarantee, testing whether disinhibition and transparency scale monotonically with guarantee strength. We additionally plant graded in-context cues within the honeypot that contradict the stated guarantee, from a faint anomaly to an explicit contradiction, and report the fracture point, the leak strength at which monitor recall returns to Arm A levels.

4.5 Part 2: Mechanistic Layer

We extract a contrastive surveillance direction, $v_{\text{surveillance}} = E[h_A] - E[h_B]$, from matched Arm A/B activations at a selected layer and token window, and test its ability to toggle obfuscation against the prompt, against matched-norm and unrelated-concept controls, with a partial toggle preregistered as the expected outcome. Given known sensitivity of single-layer steering, we pre-register simultaneous multi-layer injection rather than treating a single-layer null as refutation. The central test for our novelty claim is a cross-probe against a general evaluation-awareness direction: if $v_{\text{surveillance}}$ is not distinguishable from “am I being evaluated?” by cosine similarity and cross-transfer, our contribution reduces to a replication.

4.6 Analysis and Validity

We fit a logistic regression of monitor recall on arm, with paraphrase and problem as random effects, testing planned contrasts (A-C, B-C, A-B, B-D, A-E) under Holm-Bonferroni correction and FDR control across the layer sweep. Belief-conditioning predicts $\text{recall}_B > \text{recall}_C > \text{recall}_A$ with $B \approx D$ and A strictly above E. A no-hack-available control and a CoT-truncation check bound baseline unfaithfulness and CoT necessity as competing explanations. Before the full run, a pilot confirms a measurable base hack rate, a measurable Arm C verbalisation rate, and A/B separation. Failing any of these escalates model scale rather than proceeding on an untested premise.

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